



Repair of congenital diaphragmatic hernias on Extracorporeal Membrane Oxygenation (ECMO): Does early repair improve patient survival?

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Abstract

Introduction: The optimal timing of repair for congenital diaphragmatic hernia (CDH) patients that require ECMO is controversial. Early repair on ECMO theoretically allows for restoration of normal thoracic anatomy but entails significant bleeding risks. The purpose of this study was to examine the institutional outcomes of early CDH repair on ECMO.

Methods: The records of infants with CDH placed on ECMO from 2001 to 2011 were reviewed. Since 2009, a protocol was instituted for early repair while on ECMO. For this study, three cohorts were analyzed: early repair (<72 h), late repair (>72 h), and post-decannulation. These groups were compared for outcomes regarding morbidity and survival.

Results: Forty-six CDH patients received ECMO support with an overall survival of 53%. Twenty-nine patients (11 early/18 late) were repaired on ECMO, while 17 patients had repair post-decannulation. Survival was 73%, 50%, and 64% for those repaired early, late, or post-decannulation, respectively. Despite significantly worse prenatal factors, patients repaired early on ECMO had a similar survival. When comparing patients repaired on ECMO, the early group patients were decannulated 6 days earlier (p -value = 0.009) and had significantly lower circuit complications (p = 0.03).

Conclusion: In conclusion, early repair on ECMO was associated with decreased ECMO duration, decreased circuit complications, and a trend towards improved survival.

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Despite advances in the treatment of infants born with congenital diaphragmatic hernias (CDH), mortality rates are still significant. While some patients are born in relatively

stable physiologic condition, other high-risk infants have severe pulmonary hypoplasia and concomitant pulmonary hypertension that are beyond the treatment capabilities of conventional medical strategies [1,2]. In infants with this severe disease, extracorporeal membrane oxygenation (ECMO) has become a bridge to allow for patient stabilization [3,4].

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Utilized in 20%–40% of infants with CDH, management with ECMO has been proven to improve overall disease survival [5,6]. However, if the severity of the disease requires ECMO cannulation, mortality rates approach 50% [7,8]. One current question regarding the outcomes in high-risk infants managed with ECMO is the timing of the surgical repair in relation to ECMO utilization [3]. In theory, early repair on ECMO will normalize thoracic anatomy and relieve any mass effects from the herniated viscera, which would hopefully allow the lungs to expand and decrease pulmonary hypertension, and thus the duration of necessary ECMO use. However, as surgical repair on ECMO carries an increased risk of bleeding, others prefer to delay repair until the infant is stabilized and has either been decannulated or is approaching decannulation [9–11].

Since 2009, at our institution, we have developed an institutional protocol of CDH management that includes early CDH repair (<72 h on ECMO) for infants who are placed on ECMO. The purpose of this study was to compare infants who had undergone early repair to our historical cohort and to assess patient outcomes.

1. Methods

1.1. Patient population

From 2001 to 2011, there have been 148 patients who have presented to Texas Children's Hospital with the diagnosis of congenital diaphragmatic hernia. Fifty three patients required the use of ECMO during their hospitalization. Patients were excluded from this study if the patient died prior to operative repair. This study is a retrospective review of data that have been prospectively collected in accordance with all Health Insurance Portability and Accountability Act (HIPAA) regulations and approval by the Baylor College of Medicine Institutional Review Board (H-26176).

1.2. ECMO initiation

Since 2009, we have adopted an institutional protocol of the surgical management for CDH infants, which includes repair on ECMO within 72 h post-cannulation. ECMO was initiated when a patient was felt to be refractory to all medical therapies and in danger of impending cardiopulmonary collapse. In our institution, there are three surgeons (DLC, OOO, and TCL) who repair all neonatal CDH patients and who are part of the ECMO surgery team at Texas Children's Hospital. The decision for ECMO initiation is made between the ECMO/CDH surgeon and the ECMO neonatologist. While venovenous cannulation is considered if the internal jugular vein will permit a 13-Fr cannula, the majority of infants are placed on venoarterial ECMO (VA-ECMO), with the internal jugular vein and carotid artery cannulated via an open technique. In rare cases of hypoxemia without significant ongoing vasopressor support and an adequately sized jugular vein, venovenous cannulation is

attempted using the dual lumen bicaval cannulas; this technique of cannulation has been previously described [12].

1.3. ECMO management

Once ECMO is initiated, the target ECMO flow rate is 100 cc/kg/min and ventilation strategies are employed to minimize barotrauma. Anticoagulation is achieved by keeping the activated clotting time (ACT) at 180–220 s, anti-thrombin III levels >80%, and a heparin level between 0.3 and 0.7 U/mL. The ACT at the time of repair is kept at 160–180 s. A dose of Amicar (100 mg/kg) is given prior to the operation, and is infused for 24 h after surgery at 25 mg/kg/h. Patients are transfused when platelet levels fall below $100 \times 10^3/\text{UL}$, hematocrit drops below 30, or the INR rises above 1.5 with platelets, packed red blood cells (PRBC), and fresh frozen plasma (FFP), respectively. The type of repair was a prosthetic dual mesh Gore-Tex patch (Gore Medical, Flagstaff, AZ, USA) or a primary repair. The technique of operative repair was at the discretion of the operating surgeon and the size of the defect.

1.4. Study design

We identified three treatment groups for comparison: Early Repair (<72 h after cannulation), Late Repair (>72 h), and Post-Decannulation. We specifically compared the early repair group to both the late repair and post-decannulation repair groups. In an attempt to risk-stratify our patient population, pre-operative variables including the lung-to-head ratio (LHR) at 23 to 26 weeks gestation, observed to expected total fetal lung volume (o/e TFLV), the presence of liver herniation, the need for patch repair, gestational age, and birth weight were collected. Outcomes assessed included patient survival, the number of days spent on a ventilator and on ECMO, the number of ECMO circuit changes, bleeding complications while on ECMO, and length of hospital stay. For the purposes of comparing bleeding complications, we described those related to surgical repair on ECMO (hemothorax) separately from those related to anticoagulation (intraventricular hemorrhage).

All data are reported as a mean with a standard deviation. Patient cohorts were compared using chi-square tests, student's T-tests, and ANOVA for categorical and continuous variables. The effect of the timing of repair on survival was performed using a Kaplan–Meier analysis with log-rank test. All statistical analyses were performed using SPSS (version 19, Chicago, IL). A p-value < 0.05 was considered statistically significant.

2. Results

2.1. Patient demographics

During the study period, 53 patients underwent ECMO cannulation out of 148 total CDH patients (36%). Of those, 3

Table 1 Basic demographics and pre-operative variables.

Variable	Early Repair n = 11	Late Repair n = 18	p-value	Early Repair n = 11	Post-Decannulation, n = 17	p-value
Male	73%	67%	1.0	73%	53%	0.44
Left-side	73%	77%	1.0	73%	59%	0.69
Isolated disease	82%	83%	1.0	82%	94%	0.54
Gestational Age (mean)	37.9 (± 0.8)	37.9 (± 1.9)	0.93	37.9 (± 0.89)	37.9 (± 1.8)	0.99
Birth Weight (mean)	2705 (± 565)	2959 (± 579)	0.26	2705 (± 565)	3174 (± 570)	0.04
O/E TFLV (mean)	23% (± 5.7)	25% (± 6.7)	0.55	23% (± 5.7)	33% (± 11.7)	0.06
LHR (mean)	1.2 (± 0.32)	1.3 (± 0.26)	0.45	1.2 (± 0.32)	1.8 (± 0.77)	0.03
Liver Up	100%	89%	0.51	100%	73%	0.11

patients died immediately (<12 h of life) without repair; 4 patients died at greater than 12 h of life without repair. These patients were excluded from the subsequent analysis. Forty-six patients placed on ECMO had a repair of their diaphragm defect: 11 (24%) in the early repair group, 18 (39%) in the late repair group, and 17 (37%) in the post-decannulation groups.

The prenatal characteristics and basic demographics of the three cohorts are presented in Table 1. The early and late repair groups were similar with respect to gender, the side of the hernia, the presence of an associated disease, gestational age, and birth weight. Patients who were repaired post-decannulation had more favorable prenatal markers with a higher mean LHR and a higher O/E TFLV than the early repair patients. The percentage of patients with liver herniation into the chest was similar.

All patients were repaired via an open abdominal approach. 89% of patients required a patch repair. The number of patients receiving a patch repair was similar between all three groups (100%, 95%, 87%, respectively, $p = 0.396$). Forty patients (87%) were placed on VA-ECMO, and 6 were placed on venovenous ECMO. Three patients required conversion from VV ECMO to VA ECMO. One patient required a second run of ECMO after decannulation due to a pulmonary hypertensive crisis. There was no difference in mortality between the VV (33%) and the VA (40%) patients ($p = 1.00$).

2.2. Patient complications

When evaluating the morbidity related to the disease, the late repair group appeared to have worse outcomes despite

similar pre-operative characteristics to the early repair patients (Table 2). Late repair patients had significantly increased number of days on ECMO and total number of circuit changes. Despite worse pre-operative prognostic variables, the early repair group had similar outcomes when compared to the post-decannulation group other than a higher blood transfusion requirement ($p = 0.07$).

While there was no statistically significant difference, the late repair group had an increased number of major bleeding complications, days of intubation, and hospital days. Our bleeding complications related to ECMO use are described in Table 3. Bleeding complications were divided into medical (related to anticoagulation) and surgical (bleeding related to the surgery). Early repair on ECMO compared to post-decannulation repair did not result in statistically significant increases in rates of either medical or surgical bleeding. The most common bleeding complication was intraventricular hemorrhage, although only 3 patients had grade III or IV bleeds. The most common surgical complication was thoracic bleeding after CDH repair. Seven total patients had post-operative thoracic bleeding with three patients being in the early group. Of these, 2/7 patients required a thoracotomy and 5/7 of these patients did not survive to discharge.

2.3. Patient survival and outcomes

Overall survival of our CDH patients treated with ECMO ($n = 53$) was 53%. In patients who were able to be repaired, this improved to 61%. The survival was higher in the early repair group (73%) compared to the late repair (50%) and

Table 2 Post-operative morbidity.

Variable	Early Repair n = 11	Late Repair n = 18	p-value	Early Repair n = 11	Post-Decannulation, n = 17	p-value
Survival	73% (8/11)	50% (9/18)	0.27	73% (8/11)	64% (11/17)	1.0
Days on ECMO (mean)	12 (± 7.5)	18 (± 6.1)	0.01	12 (± 7.5)	10 (± 3.5)	0.41
Days of Intubation (mean)	42 (± 27)	67 (± 49)	0.33	42 (± 27)	38 (± 27)	0.80
Days in Hospital (mean)	110 (± 55)	144 (± 81)	0.32	110 (± 55)	64 (± 53)	0.47
Cases requiring circuit exchange	27%	72%	0.03	27%	17%	0.65
Major Surgical Bleeding Complications	36%	44%	0.72	36%	12%	0.17
Major Medical Bleeding Complications	36%	55%	0.45	36%	18%	0.38
Total Transfusion requirements (mean, cc)	712 (± 648)	803 (± 414)	0.53	712 (± 648)	323 (± 455)	0.07

Table 3 Bleeding complications related to ECMO use (all patients).

Complication	Number of patients (%) n = 46
Medical	
IVH	11 (24 %)
Cannula site bleeding	6 (13 %)
Surgical	
Thoracic bleeding	7 (15 %)
Intraperitoneal bleeding	2 (4 %)
Surgical site bleeding	3 (7 %)

post-decannulation groups (64%) ($p = 0.439$, when comparing all three groups). A Kaplan–Meier survival analysis with log rank test did not find that the timing of repair influenced the survival rates ($p = 0.345$). When comparing only patients who were repaired while on ECMO, the early repair patients appear to have more favorable outcomes compared to the late repair patients (Table 2). The early repair patients had a higher survival trend (73% vs. 50%, $p = 0.273$), fewer days spent on ECMO (12 vs. 18, $p = 0.009$), and fewer circuit exchanges (27% vs. 72%, $p = 0.03$) than late repair patients.

3. Discussion

The optimal time to perform CDH repair in infants who require ECMO is a difficult decision for the pediatric surgeon, as both early repair and late repair have significant risks. The benefits of early repair include a normalization of thoracic anatomy, a decreased mass effect of the herniated contents, and an improvement in venous return when the liver is returned to its normal position. However, bleeding risks can be substantial, and the patients may become clinically unstable during the repair [2]. Later repair, particularly after decannulation, can also add morbidity by subjecting the patient to increased time on the ECMO circuit with all of the concomitant risks of circuit complications and medical bleeding.

In this study, the overall survival in our patients is comparable to other reported survival rates in CDH patients who require ECMO [11,13]. When comparing the survival between the early repair group and the other two cohorts, we were unable to find a statistically significant difference amongst the three groups. One interesting point is that even though the early repair group appeared to be a more severe CDH cohort by prenatal imaging, the survival was equal between the early repair group and the post-decannulation group. Furthermore, for patients repaired on ECMO, the early repair group had a 23% higher survival than the late repair group. In reviewing the literature, these results are almost identical to the paper by Dassinger et al. where their survival was 71% with an average time to repair on ECMO of 21 h [14]. Additionally, one of the significant benefits was

that the early repair group spent almost 1 week less time on ECMO and had a significantly lower number of circuit complications. The use of blood products was also lower in the early repair group, which is most likely associated with the decreased time on ECMO.

With respect to operative complications, we were unable to appreciate a difference in major bleeding complications between the early repair group and the other two cohorts. One of the arguments against early repair on ECMO has been the presumption that a significant bleeding risk may lead to increased patient mortality [15]. In this study, we did not observe this result; one possible explanation for this observation may be a more sophisticated monitoring of the patient's anti-coagulation profile at our institution. The role of comprehensive anti-coagulation management for ECMO patients, beyond just the ACT, has been supported by a growing body of literature [16–18]. Khaja et al. summarized within their study the multi-factorial nature of ECMO anticoagulation, concluding that a host of coagulation parameters needs to be monitored and titrated for optimal outcomes [16]. Since 2006, the use of an expanded ECMO coagulation profile was implemented at our institution, encompassing an array of coagulation measures such as partial thromboplastin time, fibrinogen, activated clotting time, prothrombin time, heparin levels, and anti-thrombin III levels. This comprehensive monitoring may explain why the incidence of bleeding was not higher in the early repair group and why operating on ECMO has become safer than historically accepted [15]. However, the ramifications of operative bleeding on ECMO should not be minimized, as 5 out of the 7 patients with thoracic bleeding (in both early and late groups) did not survive in the post-operative period.

To date, the evidence supporting one management strategy over another is not conclusive [3]. A recent study by Dassinger et al. compared their institutional outcomes to the CDH registry outcomes and found that the ECMO patients repaired at the institution were repaired earlier (2.3 days vs. 6.9 days) with a higher survival rate (71% vs. 49%, $p = 0.016$) compared to the registry patients [14]. Their conclusion from that study was that early repair can be accomplished with acceptable morbidity and mortality. A study by Downard et al. found a possible benefit in using aminocaproic acid during CDH repair on ECMO, as reoperation rates decreased by 21% when comparing a non-Amicar historical group to Amicar patients in the ELSO database [19]. In contrast, a study by Bryner et al., also utilizing the CDH registry, found that repair after cessation of ECMO therapy was associated with improved survival [11].

Of note, this current study does have a number of limitations, such as the small sample size and the duration of the study period. Unquantifiable heterogeneity due to changing practices in ECMO management with regards to anticoagulation schemes and ventilation strategies could confound our results; however it is conceivable that the evolution of these practices such as anticoagulation management has allowed for safer surgery while on

ECMO. Even though this study has a number of limitations, the current body of literature only addresses the safety of diaphragm hernia surgery pre-ECMO versus post-ECMO [2,11,14,18,20]. This study is the first to address the timing of repair on ECMO, presenting a direct comparison between an early versus later repair cohort on ECMO.

In conclusion, the direction of this study seems to preliminarily demonstrate that early repair on ECMO does not pose a higher risk for mortality for CDH patients, and may be beneficial with regards to outcomes concerning decreased time on ECMO and ECMO morbidity. Future studies are needed in larger patient samples to prospectively compare the technique of early repair of congenital diaphragmatic hernia on ECMO to more traditional management strategies.

References

- [1] The Congenital Diaphragmatic Hernia Study Group. . Estimating disease severity of congenital diaphragmatic hernia in the first 5 minutes of life. *J Pediatr Surg* 2001;36(1):141-5.
- [2] Lally KP, Paranka MS, Roden J, et al. Congenital diaphragmatic hernia. Stabilization and repair on ECMO. *Ann Surg* 1992;216(5):569-73.
- [3] Khan AM, Lally KP. The role of extracorporeal membrane oxygenation in the management of infants with congenital diaphragmatic hernia. *Semin Perinatol* 2005;29(2):118-22.
- [4] Harrington KP, Goldman AP. The role of extracorporeal membrane oxygenation in congenital diaphragmatic hernia. *Semin Pediatr Surg* 2005;14(1):72-6.
- [5] Atkinson JB, Ford EG, Humphries B, et al. The impact of extracorporeal membrane support in the treatment of congenital diaphragmatic hernia. *J Pediatr Surg* 1991;26(7):791-3.
- [6] Morini F, Goldman A, Pierro A. Extracorporeal membrane oxygenation in infants with congenital diaphragmatic hernia: a systematic review of the evidence. *Eur J Pediatr Surg* 2006;16(6):385-91.
- [7] Hoffman SB, Massaro AN, Gingalewski C, et al. Predictors of survival in congenital diaphragmatic hernia patients requiring extracorporeal membrane oxygenation: CNMC 15-year experience. *J Perinatol* 2010;30(8):546-52.
- [8] Guner YS, Khemani RG, Qureshi FG, et al. Outcome analysis of neonates with congenital diaphragmatic hernia treated with venovenous vs venoarterial extracorporeal membrane oxygenation. *J Pediatr Surg* 2009;44(9):1691-701.
- [9] Rozmiarek AJ, Qureshi FG, Cassidy L, et al. Factors influencing survival in newborns with congenital diaphragmatic hernia: the relative role of timing of surgery. *J Pediatr Surg* 2004;39(6):821-4 [discussion 821-4].
- [10] Reyes C, Chang LK, Waffarn F, et al. Delayed repair of congenital diaphragmatic hernia with early high-frequency oscillatory ventilation during preoperative stabilization. *J Pediatr Surg* 1998;33(7):1010-4 [discussion 1014-6].
- [11] Bryner BS, West BT, Hirschl RB, et al. Congenital diaphragmatic hernia requiring extracorporeal membrane oxygenation: does timing of repair matter? *J Pediatr Surg* 2009;44(6):1165-71 [discussion 1171-2].
- [12] Lazar DA, Cass DL, Olutoye OO, et al. Venovenous cannulation for extracorporeal membrane oxygenation using a bicaval dual-lumen catheter in neonates. *J Pediatr Surg* 2012;47(2):430-4.
- [13] Tiruvoipati R, Vinogradova Y, Faulkner G, et al. Predictors of outcome in patients with congenital diaphragmatic hernia requiring extracorporeal membrane oxygenation. *J Pediatr Surg* 2007;42(8):1345-50.
- [14] Dassinger MS, Copeland DR, Gossett J, et al. Early repair of congenital diaphragmatic hernia on extracorporeal membrane oxygenation. *J Pediatr Surg* 2010;45(4):693-7.
- [15] Vazquez WD, Cheu HW. Hemorrhagic complications and repair of congenital diaphragmatic hernias: does timing of the repair make a difference? Data from the Extracorporeal Life Support Organization. *J Pediatr Surg* 1994;29(8):1002-5 [discussion 1005-6].
- [16] Khaja WA, Bilen O, Lukner RB, et al. Evaluation of heparin assay for coagulation management in newborns undergoing ECMO. *Am J Clin Pathol* 2010;134(6):950-4.
- [17] Baird CW, Zurakowski D, Robinson B, et al. Anticoagulation and pediatric extracorporeal membrane oxygenation: impact of activated clotting time and heparin dose on survival. *Ann Thorac Surg* 2007;83(3):912-9 [discussion 919-20].
- [18] Keijzer R, Wilschut DE, Houmes RJ, et al. Congenital diaphragmatic hernia: to repair on or off extracorporeal membrane oxygenation? *J Pediatr Surg* 2012;47(4):631-6.
- [19] Downard CD, Betit P, Chang RW, et al. Impact of AMICAR on hemorrhagic complications of ECMO: a ten-year review. *J Pediatr Surg* 2003;38(8):1212-6.
- [20] Davis PJ, Firmin RK, Manktelow B, et al. Long-term outcome following extracorporeal membrane oxygenation for congenital diaphragmatic hernia: the UK experience. *J Pediatr* 2004;144(3):309-15.